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192524185 CSA155: Cloud Computing.

1. Cloud computing - 4 characteristics

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, & software over the internet.

It allows users to access resources whenever needed without purchasing & maintaining their own physical infrastructure.

4. Characteristics:

1. On-demand Self-service:

Users can access computing resources whenever required for the maintanence.

ex: Creating a virtual server through AWS

2. Elasticity:

The Elasticity is the term where the resources can be increased or decreased easily.

Ex:

3. Broad Network Access:

Cloud computing can be accessed through the internet

ex:

4. Resource Pooling.

Computing resources can be increased or decreasing.

Ex: Multiple organizations using the same cloud.

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2. Trace the evolution of cloud computing. Difference from :

1. Mainframe computers

The use of mainframe computers in early period large centralized computers served many users through terminals.

2. Traditional Web Services :

The web started with the version ~~wb~~ web 1.0, Web 2.0, web 3.0, and goes on.

3. Personal computers & client server computing

It became widespread. Client server architecture allowed computers to request services.

4 Virtualization

One physical server could run multiple virtual machines. This improved hardware utilization & allowed flexible resources.

5. Modern Cloud Platforms.

High speed Internet enabled providers to offer scalable resources as services.

Software evolution : Web applications & APIs enabled Internet over resources.

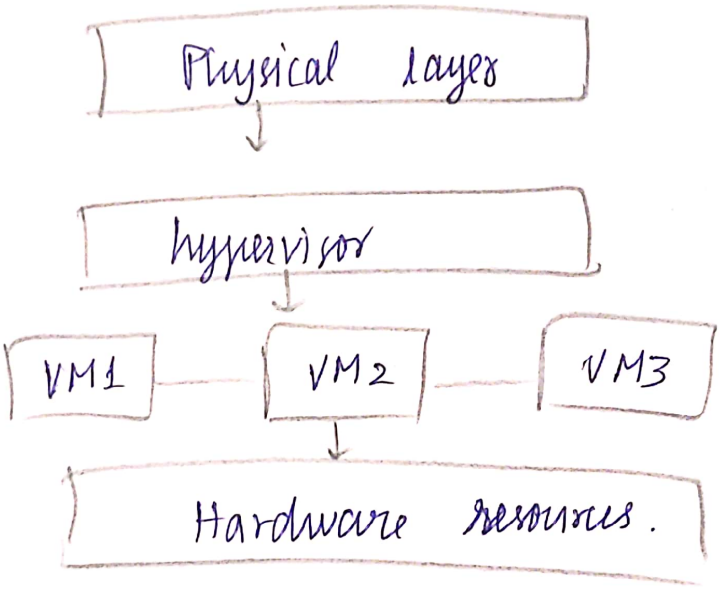
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Server Virtualization.

Server virtualization is a technology that divides one physical server into multiple resources.

ex: VM

A software layer called a hypervisor creates and manages these virtual machines.



Virtualization

A technology that creates virtual versions of computing resources.

Cloud computing.

A service model that delivers computing resources

4

A data center is a physical facility that contains servers, storage systems, networking equipment & supporting infrastructures.

Main components.

1. Servers
2. Storage system
3. Networking equipments.
4. Power & cooling systems.
5. Security & Monitoring

1. Data Storage :

5

Cloud service

1. IaaS - Infrastructure as a service
   Ex: Amazon - EC2
2. PaaS - Platform as a service
   Ex. Google app Engine
3. SaaS - Software as a Service
   Google docs.
4. XaaS - Anything as a service.
   Amazon RDS.

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